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DevAIOps: A Systematic Literature Review on Integrating Explainable AI Agents into CI/CD Pipelines for Optimization, Security, and Auto-Remediation

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Abstract

This paper presents a systematic literature review (SLR) on DevAIOps — the integration of explainable AI agents and Large Language Models (LLMs) into DevOps and DevSecOps workflows. Following PRISMA, 222 candidate publications from Scopus and Web of Science and manually identified seminal works were screened to 217 unique records, 47 eligible studies, and ten high-impact studies (2019–2025) synthesized in depth across AIOps, CI/CD optimization, DevSecOps, explainable AI, and LLM agents for autonomous remediation. Guided by three research questions on AI techniques, explainability, and security in DevOps automation, the synthesis reveals four critical gaps: the absence of a unified DevAIOps architecture, the absence of standardized KPI benchmarks (notably MTTR), the lack of agent-level explainability adapted to pipelines, and the under-studied risk of LLM hallucinations in autonomous remediation aligned with the OWASP Top 10 for LLM and for Agentic AI Applications (2026). For the two trust gaps, a trust loop is proposed in which artifact-linked explanations feed OWASP-aligned multi-layer validation and risk-tiered human-in-the-loop gating before any remediation is applied. For each gap, concrete research directions are formulated toward trustworthy, explainable, and secure DevAIOps.

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Peer-review under responsibility of the scientific committee of the technical program committee members.

Keywords: DevAIOps; AIOps; CI/CD pipeline; explainable AI; LLM agents; auto-remediation; DevSecOps; OWASP; LLM hallucination; MTTR; systematic literature review; PRISMA

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1. Introduction

Today, software development requires CI/CD pipelines whose growing complexity exceeds what humans can manage manually. The reasons for failure are hidden in massive logs, monitoring data overwhelms the operators, and automatically generated advice by an AI solution needs to be validated before deployment. Artificial Intelligence for IT Operations (AIOps) has been introduced to overcome those challenges by using ML/DL and LLMs to analyze operational data – logs, metrics, traces, and build artifacts – to detect, diagnose, and remediating problems. AIOps was formally introduced by Dang et al. [2] at ICSE 2019, with alert fatigue and the shortage of labeled datasets being identified as two major bottlenecks for industry-level adoption, which were addressed only partially. Nonetheless, the demand stays strong: the global market for generative AI in DevOps amounted to USD 942.5 million in 2022, and will hit USD 22.1 billion by 2032 (38.2% CAGR) [11]. However, embedding autonomous AI agents in vital pipelines comes along with the new challenges – hallucination in LLM-based remediation recommendations, non-transparency of automated decisions, and lack of frameworks for security of agent's behavior.

In this paper we introduce the term DevAIOps to refer to the upcoming paradigm of explainable and controllable AI agents in DevOps/DevSecOps workflows, resulting in changes of KPIs such as MTTR, deployment success rate, and security compliance.

The relevance of the topic lies in the fact that mobile and pervasive computing rely on CI/CD pipelines for the continuous delivery of mobile apps (supported by Fastlane, Bitrise, and GitHub Actions), GitOps and Kubernetes for over-the-air and edge deployments, and AIOps for the monitoring of distributed mobile backends and fleet of devices. Thus, reliable and secure pipeline automation becomes the foundation for dependable mobile and pervasive systems. Three research questions (RQs) drive the review:

RQ1. What AI, ML, and LLM techniques are used in CI/CD pipelines and AIOps?

RQ2. How does explainability get addressed in DevOps automation driven by AI?

RQ3. What are the security risks, including LLM hallucinations, associated with LLM-based autonomous remediation and how are they mitigated?

To answer the RQs, this paper provides the methodology of a reproducible review, tracking all searches, scoring, and screening steps over 217 unique papers; thematic taxonomy over five dimensions; gap analysis; and a research agenda, mapping each gap into concrete research directions. While the first three items constitute the scope of this paper, the last item constitutes the main contributions of the paper. Section 2 describes the methodology of the review; Section 3 summarizes results in five themes; Section 4 presents the gap analysis and research directions; Section 5 describes the threats to validity; and Section 6 concludes. Figure 1 shows the reference DevAIOps architecture motivating the review.

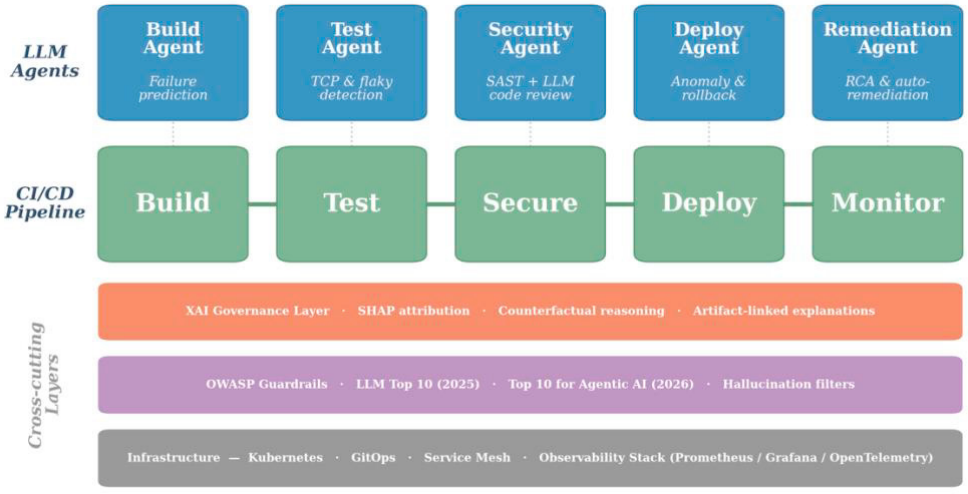


Fig. 1. DevAIOps reference architecture concept

2. Review Methodology

The systematic review conducted is done using the PRISMA framework for systematic literature review in software engineering, including five stages described in Figure 2. The process for each stage was carefully scripted to ensure that it was repeatable; the complete list of all 47 articles that were used in the review can be obtained from the authors on request.

2.1. Search Strategy and Selection Criteria

The literature search involved two major scientific databases, Scopus and Web of Science, as well as a select number of manually sourced seminal papers, following the recommendations of the PRISMA guidelines on sources other than database searches. The search terms involved combining the major domain terms with minor technique terms (Table 1) within five thematic domains (T1-T5): T1 - AIOps/automated IT operations; T2 - CI/CD pipeline optimization using AI; T3 - DevSecOps/AI-powered security; T4 - XAI/explainable AI in software operations; and T5 - LLM agents/large language model agents for autonomous software operations.

Table 1. Search protocol parameters.

Parameter	Value
Sources	Scopus; Web of Science; manual search (seminal works)
Date range	January 2019 – April 2026
Query date	19 April 2026
Language	English only
Fields searched	Title, abstract, keywords
Primary terms	"AIOps", "DevAIOps", "AI-driven DevOps", "CI/CD optimization", "DevSecOps"
Secondary terms	"LLM agent", "large language model", "explainable AI", "XAI", "auto-remediation", "root cause analysis", "anomaly detection", "hallucination"
Example query	("AIOps" OR "AI-driven DevOps" OR "CI/CD") AND ("large language model" OR "LLM agent" OR "explainable AI" OR "auto-remediation")

Studies were retained if they (i) address the intersection of AI/ML/LLM and DevOps, DevSecOps, MLOps, or AIOps in a CI/CD or software-operations context; (ii) are published in a peer-reviewed venue indexed in Scopus or Web of Science; and (iii) contribute empirical evidence, a systematic survey, a novel framework, or a production-validated case study. Exclusion was coded E1 — wrong application domain (IoT, medical, energy, autonomous driving, aviation, finance, chemistry) even where AI techniques are used — or E2 — absence of an AI component or of a DevOps/CI/CD context.

2.2. Relevance Scoring Formula

To rank the deduplicated corpus objectively, a composite relevance score (0–100) was computed for every paper using a scripted formula aggregating the seven weighted components detailed in Table 2. Tier 1 Keywords are those pertaining to DevOps and AIOps concepts, whereas Tier 2 Keywords refer to the names of particular AI/ML methods. Formally speaking, the score of the relevance for a paper p is calculated according to the following formula:

$$Score(p) = C_{cit} + C_{inf} + V + R + O + K_1 + K_2 \qquad (1)$$

where every addend represents the corresponding capped score of the component from Table 2. Such an aggregated score serves as a custom-made ranking metric used exclusively for ordering the corpus in the context of research papers selection. If there were duplicated papers after deduplication process, the one with more complete metadata (i.e., abstract and higher citations count) was chosen.

Table 2. Composite relevance score (maximum 100 points)

Component	Max	Formula / Basis	Coefficient rationale
Citation impact (C_{cit})	15	$\min(15, \log_{10}(\text{citations} + 1) \times 8)$	Log transformation limits the effect of outliers; roughly 300 citations represent an eight times multiplier.
Influential citations (C_{inf})	8	$\min(8, \text{influential_citations} \times 1.0)$	One signal-high citation represents one point, and eight citations result in saturation.
Venue quality (V)	12	$\min(12, \text{venue_tier_score} \times 0.6)$	Rescale from 0-20 venues to 0-12 range.
Recency (R)	10	$\max(0, \min(10, (\text{year} - 2019) \times 10/7))$	Linearly map 2019-2026 year period to 0-10 interval.
Open access (O)	5	5 if open-access PDF available, else 0	Give a constant bonus if the source is open access.
Tier-1 keywords (K_1)	25	$\min(25, \text{tier1_hits} \times 5.0)$	Multiply by 5.0 the domain anchors; five hits give the maximum points for this component.
Tier-2 keywords (K_2)	25	$\min(25, \text{tier2_hits} \times 2.5)$	Multiply by 2.5 the technique terms, equivalent to half the weight of the Tier-1 category.

2.3. *Quality Assessment and Study Selection*

For each paper included in the synthesis, a total of five quality criteria were applied. These were derived from existing quality criteria for systematic literature review in software engineering and included scoring for each criterion on an absolute scale ranging from completely fulfilled (1.0), partially fulfilled (0.5) to not fulfilled (0.0). Each study could thus obtain a quality score up to 5.0. Criteria QA are as follows: QA1 – Are the research questions well-defined? QA2 – Is the methodology well-defined? QA3 – Is the study empirically/experimentally validated? QA4 – Are the results clearly stated and quantified? QA5 – Is the work presented in a peer-reviewed journal/conference?

A full score of 4.0/5.0 or better was obtained by all ten studies; seven studies received the maximum score, [7] had a score of 4.5 (QA3 partially fulfilled), while [2] and [10] got 4.0, QA3 and QA4 being partially fulfilled due to their nature as surveys rather than primary empirical studies.

The selection funnel—showing how the number of records was reduced from 222 to 47 relevant studies—is shown in Fig. 2 below.

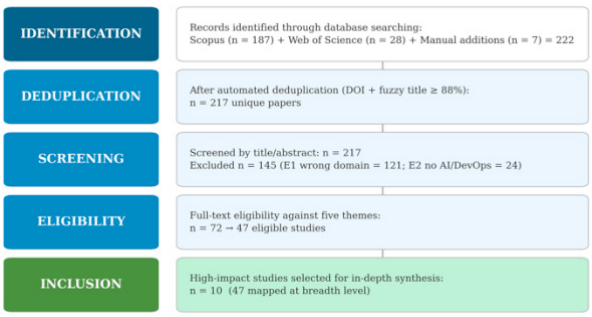


Fig. 2. PRISMA flow diagram for the DevAIOps SLR

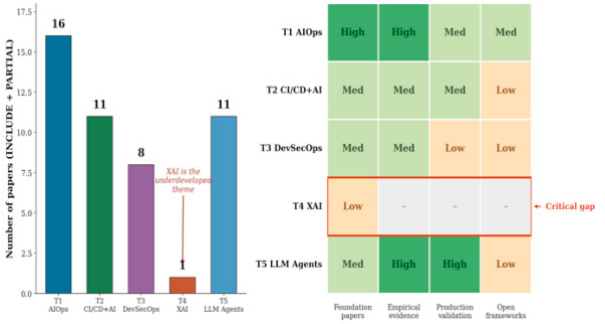


Fig. 3. Theme distribution across in the 47-study corpus (left) and maturity heatmap by theme and dimension (right).

Out of the 47 included studies, ten have been selected for a more detailed qualitative synthesis in an intentional two-tier approach, not as a constraint but as an advantage: the former provides breadth (Table 3), whereas the latter provides depth within the boundaries imposed by the conference page limit. The selection was made out of the best-rated papers using the scoring scheme introduced in Table 2 and was deliberately diversified among all five themes (T1-T5). In particular, the most important CI/CD paper (T2, eleventh ranked) and the landmark DevSecOps survey [10] (lower rank because of the paper's recent publication date and secondary status) were kept in order to provide coverage for all five themes. With the repetition of findings within themes, thematic saturation is reached, and the remaining papers are summarized in terms of breadth.

3. Synthesis of Findings

This part summarizes the ten in-depth reviews in each of the five topics (T1-T5). The whole corpus of 47 studies is mapped on the basis of its breadth in Table 3. Figure 3 (presented above in Section 2.3 along with the PRISMA figure) shows the distribution of the corpus by the five topics through a maturity heatmap: T1 (AIOps), T3 (DevSecOps), and T5 (LLM agents) have higher maturity, while T4 (XAI in DevOps) demonstrates significant immaturity. Table 3 lists the top fifteen studies of the whole corpus according to their ranking by topic, AI approach, DevOps stage, security focus, explainability capability, and evaluation type; the ranking of all 47 studies can be obtained from the authors.

Table 3. Top 15 of the 47 studies that have been considered eligible, arranged by descending composite relevance score (Score, a specially designed metric defined in Section 2.2 and used only for ranking purposes). Abbreviations: Th = topic; Sec = security focus; XAI = explainability support; Eval = evaluation type.

Rank	Title	Yr	Th	AI Method	DevOps Stage	Sec	XAI	Eval	Score
1	A Survey on Automated Log Analysis for Reliability Engineering	2021	T1	Survey	Operate	–	–	Secondary	81
2	AIOps: Real-World Challenges and Research Innovations	2019	T1	ML	Cross	–	–	Conceptual	80.9
3	Automated Root Causing of Cloud Incidents using In-Context Learning with GPT-4	2024	T5	LLM	Operate	–	–	Empirical	75.4
4	Anomaly Detection and Failure Root Cause Analysis in (Micro)Service-Based Cloud Applications: A Survey	2023	T1	Survey	Operate	–	–	Secondary	66.2
5	Self Adjusting Log Observability for Cloud Native Applications	2024	T1	ML	Deploy	–	–	Empirical	62.6
6	Learning Representations on Logs for AIOps	2023	T1	Deep Learning	Operate	–	–	Empirical	62.4
7	X-Lifecycle Learning for Cloud Incident Management using LLMs	2024	T5	LLM	Operate	–	–	Empirical	61.9
8	Assess and Summarize: Improve Outage Understanding with Large Language Models	2023	T5	LLM	Operate	–	–	Empirical	60.7
9	Exploring LLM-Based Agents for Root Cause Analysis	2024	T5	LLM	Operate	–	–	Empirical	60.1
10	Self-supervised anomaly detection from distributed traces	2020	T3	LLM	Secure	Yes	Yes	Empirical	58.5
11	LogSage: An LLM-Based Framework for CI/CD Failure Detection and Remediation with Industrial Validation	2025	T2	LLM	Build	–	Yes	Empirical	57.3
12	LogExpert: Log-based Recommended Resolutions Generation using Large Language Model	2024	T1	LLM	Build	–	–	Empirical	56.4
13	Comparing Transformer-Based Log Anomaly Detection with LLM Explainability in CI/CD Pipelines	2025	T4	LLM	Deploy	–	Yes	Empirical	56.3
14	Augmenting Automatic Root-Cause Identification with Incident Alerts Using LLM	2024	T5	LLM	Operate	–	–	Empirical	55.8
15	A Scalable Orchestration Architecture for Managing AI Models at Scale on Kubernetes Platforms	2025	T1	Classical ML	Deploy	–	–	Conceptual	54.8

The ten thorough articles (references [1] to [10]) are distributed in the whole ranked set; the full ranking of 47 articles is available from the authors on request. The corpus mainly comprises empirical researches (61%), while the share of secondary and conceptual literature is 26% and 13%, respectively.

3.1. AIOps and Automated IT Operations (T1)

According to Dang et al. (2), AIOps can be described as machine learning for monitoring, event correlation, root cause analysis (RCA), and automated remediation. The taxonomy offered by Soldani and Brogi (4) presents the most thorough classification of anomaly detection and root cause analysis in microservices where multi-modal techniques always exhibit better performance than single-source techniques. He et al. (1) emphasize log analysis as a key component of AIOps, which includes parsing, compression, anomaly detection, and fault diagnosis and demonstrate the dominance of deep learning techniques (LSTM, Transformer, BERT) over statistical techniques as well as the absence of an end-to-end system that combines all these techniques.

3.2. AI-Driven CI/CD Pipeline Optimization (T2)

According to Xu et al. [5], LogSage is the most representative current implementation of DevAIOps that combines end-to-end, large language model-based framework of RCA and automated fixing of CI/CD issues using industrial data. This solution incorporates token efficient log pre-processing, structured diagnosis prompting, and retrieval-

enhanced remediation and reaches 82% failure classification accuracy, 74% RCA accuracy, and 61% acceptance rate of developer fixes. However, the method is limited to one environment, does not provide any insight into the decision-making process in diagnostics, and ignores cascaded failures in microservices. It is because of these restrictions that the following extensions were developed.

3.3. DevSecOps and AI Security Integration (T3)

Most detailed analysis of the implementation of artificial intelligence in DevSecOps has been made in the paper by Fu et al. [10]. According to them, the main problem faced in this area of research is the absence of AI systems that would cover all seven steps of the DevSecOps pipeline. They provide an example of how useful machine learning can be for SAST with a decrease in the rate of false positives of the order of 40%-60%, how useful LLMs can be for the review of code with the detection of 73% of the OWASP Top 10 vulnerabilities, and how useful ML-based anomaly detection in runtime is in reducing incident response time by 65%.

3.4. Explainable AI in Software Operations (T4)

It becomes essential for engineers to use, assess and intervene in the process of decisions taken through automation technology because of its explainability. Moreover, explainability is also the most underdeveloped area in this context since no single paper among those ranked highly emphasizes it as the main contribution, which is interesting on its own. There exist established frameworks for XAI, such as SHAP, LIME, Counterfactual, and attention-based approaches, which have already been extensively researched in the context of cybersecurity and IoT but remain unexplored in terms of application to automated processes used for developing CI/CD pipelines.

3.5. LLM Agents for Autonomous Software Operations (T5)

According to Zhang et al. [3], in-context learning GPT-4 performs 35% better in accuracy compared to rules-based approaches for cloud incidents RCA, and the MTTR for RCA is reduced by 42% in production. The X-Lifecycle framework by Goel et al. [6] that uses cross-phase attention to combine the data from different SDLC phases decreases the false positives by 28% and MTTR end-to-end by 31%. Jin et al. [8] introduce Assess-and-Summarize pipeline with chain-of-thought prompting that yields 78% of outage summaries helpful according to on-call engineers. Roy et al. [9] use ReAct-type agent with five tools to reach 68% accuracy for RCA among top-1 answers and 45% for non-agent based methods; removal of any of those five tools causes 12–20% drop in accuracy.

4. Research Gaps and Directions

In the comparative analysis of the literature, there emerge four main gaps. Two of these gaps are fundamental: a unified architecture gap (Gap 1), as well as the absence of benchmarks for the key performance indicators (Gap 2), which are presented concisely below. Finally, the other two gaps have to do with trust: agent-level explainability (Gap 3, RQ2), as well as hallucination-related security risks (Gap 4, RQ3). It is in these gaps that the body of research is most deficient.

4.1. Foundational Gaps — Unified Architecture and KPI Benchmarks (Gaps 1 and 2)

Gap 1 - Unified architecture: No existing literature combines pipeline optimization, security, and explainability. The most relevant competitor, LogSage [5], incorporates none of these dimensions; past literature has addressed all of these aspects separately [1][2][4][10], and LLM agent frameworks [3][6][8][9] have focused solely on RCA, without incorporating CI/CD. **Research direction:** a DevAIOps platform open source framework that utilizes specialized agents for build, security, and remediation processes, all governed under one explainability layer.

Gap 2 - Standard KPI benchmarks: MTTR reduction reported in the studies (31–55% improvements within Microsoft studies [3][6][8]) is limited by proprietary infrastructure and not externally verifiable, and no standard benchmarks exist for false fix ratio, OWASP compliance, or explainability coverage. **Research direction:** a multi-

dimensional KPI set (MTTR, false fix ratio, test coverage improvement, deployment success rate, security compliance) based on replicable process and demonstrated over at least three open source projects.

4.2. Gap 3 — Agent-Level Explainability for Pipeline Decisions

The current solutions for XAI deal only with the model level of explainability (the reasoning behind the rollback action by the agent and the artifacts that enabled that) rather than at the agent level. Although Roy et al. [9] come close to this issue by using the ReAct traces, their traces consist of unstructured text which is not auditable and artifact-aware explanations.

Research direction: design a pipeline native XAI module for CI/CD agents producing artifact-linked explanations. This involves augmenting the existing ReAct trace description [9] with references to artifacts and introducing SHAP attribution over the agent's input.

4.3. Gap 4 — LLM Hallucination Risk in Autonomous Remediation

The DevSecOps ecosystem recognizes the issue of hallucination; that is, LLM-generated codes are flawed because of security vulnerabilities but their effects on the CI/CD auto-remediation process have not been studied. No research addresses the question of how often these remediations are wrong or proposes an architecture for verifying the fixes to the pipelines. This need arises from the upcoming 2026 OWASP Top 10 for Agentic AI Applications with issues such as violations of autonomous actions, cascading agent failures, and unauthorized execution caused by incorrect production fixes.

Research direction: measure the hallucinations rates for advanced LLMs (such as GPT-5, Llama 4 Maverick, Mistral Large 3) on real CI/CD failure datasets, and create an architecture for validating those fixes using static analysis, OWASP Top 10 for LLM applications, OWASP Top 10 for Agentic AI Applications (2026) and progressive human-in-the-loop gates. Figure 4 illustrates the trust loop between explainability and validation.

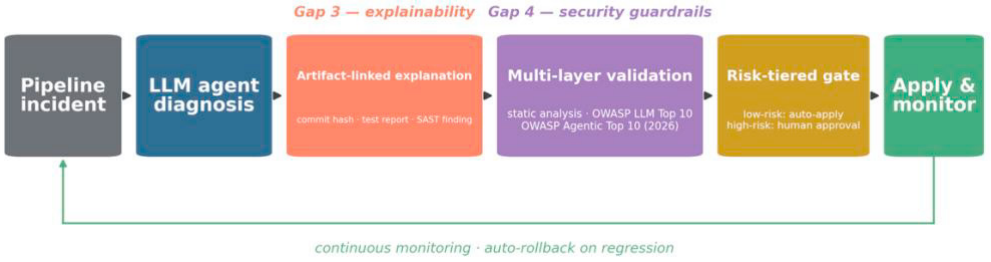


Fig. 4. Suggested trust loop for the autonomous remediation: artifact-linked explanations (Gap 3) provide input to multi-layered OWASP-based validation and risk-tiered gating (Gap 4).

5. Threats to Validity

A number of potential validity threats for this SLR are discussed below in line with SLR standards.

Construct validity: Keyword search might have missed some studies that utilized terminology beyond the predefined keyword set (keyword bias). This issue has been overcome to some extent due to the use of both primary and secondary terms in five major themes but terminological bias still exists.

Internal validity: Study selection has been carried out based on a combination of a scoring algorithm and single-reviewer screening. Possible methods to overcome this issue involve programming the algorithm and deduplication of records to achieve deterministic ranking and record decision-making. However, single-reviewer screening is still considered a threat compared to multiple reviewers.

External validity: Searching has involved mostly Scopus and Web of Science database searches and a few seminal papers. Works that were only available in arXiv, non-indexed, and grey literature could have been left out (database-coverage bias), and thus indexing quality was preferred to comprehensive coverage.

Conclusion validity: All quantitative information (like mean time to repair decreases) is provided as it has been stated by previous researchers. Publication bias may lead to overrepresentation of positive findings; the literature after 2025–2026 is under-represented, and maturity goals are stated as research objectives.

6. Conclusion

This paper presented a systematic literature review on the intersection of AI, DevOps, security, and explainability — the space defined here as DevAIOps — synthesizing ten high-impact studies in depth and mapping 47 eligible studies under a reproducible PRISMA protocol. A central conclusion is that explainability is consistently neglected across the corpus: even the strongest CI/CD-optimization frameworks (T2) provide no explanation of their automated decisions, and explainability as a dedicated theme (T4) is the least-developed area of the entire review, while the security of autonomous, agentic remediation (T3) is acknowledged but remains under-addressed.

In response to the research questions, AI techniques in CI/CD and AIOps have matured from classical machine learning toward LLM-based agents, with log analysis, anomaly detection, and root-cause analysis the most developed areas [1][2][4][7] (RQ1); explainability for agents operating inside CI/CD pipelines is largely absent despite mature XAI in adjacent domains (RQ2); and LLM-based autonomous remediation introduces security risks — notably hallucinated actions — that the OWASP Top 10 for Agentic AI Applications (2026) provides a timely framework to mitigate (RQ3). The findings identify four major gaps which need to be explored in future research, namely: (1) The need for an overarching DevAIOps framework, (2) benchmarking of KPIs, (3) Explainability at the agent level, and (4) quantification of Hallucination risk. Further work on these topics will entail the creation of the reference platform shown in Figure 1.

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